



Windows 11 gaming on Snapdragon X2 Elite is pretty good

At the Snapdragon X2 Elite architecture deep dive, I was able to get my hands on engineering samples of Windows 11 laptops running on Qualcomm's upcoming AI PC chip – specifically the 18-core Snapdragon X2 Elite Extreme variant, with 5-GHz of single core performance. Here are some observations on gaming benchmarks I ran and how the Adreno X2 GPU – the main gaming engine inside the Snapdragon X2 Elite (Extreme) – performed during these sessions.

1) Black Myth: Wukong: On the 18-core Qualcomm Snapdragon X2 Elite Extreme laptops running on 32GB of RAM and Windows 11 Enterprise, the onboard Adreno X2 GPU came out with flying colours, handling Wukong's gameplay with relative ease. I noticed frame rates never dropped below 60 frames at any given time of the game run, with Black Myth Wukong hovering between 76-82 fps for almost of the 10 mins I played the game.

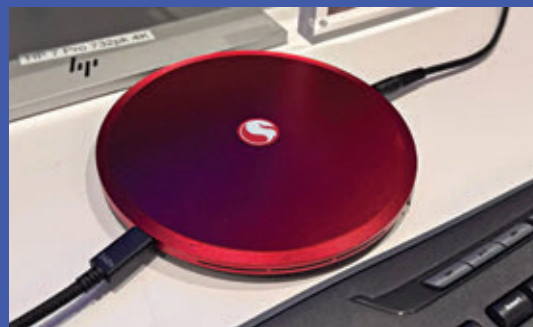
2) Cyberpunk 2077: While playing this game for a good 5 mins or so, again I noticed the frame rate never fell below 60 frames throughout – but it never crossed 70 fps either. I recorded a steady 63-68 fps average performance on the Cyberpunk 2077, with the Adreno X2 GPU on the Snapdragon X2 Elite Extreme delivering very good ray tracing performance. The test laptop didn't overheat along its edges or the keyboard, which is good to know.

3) Overwatch 2: Compared to Black Myth Wukong and Cyberpunk 2077, it has a significantly lower graphical load on the GPU. Frame rates were in line with this expectation, minimum I saw was 97-98 fps with average around 105-110 fps, with peaks of 120 fps on Overwatch 2 gameplay run. I didn't encounter any jaggies or motion blur of any kind during the fast-paced manoeuvring of the in-game gun and while quickly switching directions. The Adreno X2 GPU handled this game with a yawn to be honest, without breaking any sweat!

Qualcomm's super slim AI PC concepts

Hear the word "PC" and it's almost never associated with anything "cool" or "interesting." When was the last time you truly gasped at a PC? I don't think I have in a long time. But that certainly changed after seeing what Qualcomm's cooking up inside their secretive product design and engineering labs. What I saw didn't as much make me gasp in astonishment as it made me sit up and take notice after a long day spent in technical deep dive of the Snapdragon X2 Elite platform.

1) CD-cover shaped AI PC: The first reference design that I found quite interesting from Qualcomm is what it calls the Longcheer Mini PC. I call it the CD-cover shaped PC. This circular disc shaped AI PC runs a 18-core Snapdragon X2 Elite



chip, has 32GB of RAM and 1TB of storage. The whole thing couldn't have been thicker than 1 centimetre or 10 mm, it was that thin.

It was running Windows 11 smoothly, connected to a HP 7 Pro 4K display, and even handled a Topaz Labs AI upscale task for a small 10-second video clip – all without breaking a sweat or heating up externally on the touch.

2) Floppy disk shaped AI PC: The second most interesting reference design I saw inside Qualcomm's engineering labs is the All-in-one PC version of the Longcheer Mini PC. It resembled a 3.5-inch floppy disk, something I grew up using with desktop PCs in the 1990s and early 2000s.

This floppy disk-shaped AI PC was also running on an 18-core Snapdragon X2 Elite chip, supported by 32GB RAM and 1TB storage. It plugged into a compatible 32-inch 4K display, thus becoming a combo all-in-one Copilot+ PC.